

A Comparison of Most Popular Processors Among OEMs for Various Embedded Applications

Applications	Module	Clock Speed	Cores	Threads	Co-processor	Architecture
PC compatible computer on modules, fanless box PCs	AMD G T16R	0.62GHz	1	1	GPU AMD Radeon HD 6250 (276MHz)	x86
PC compatible computer on modules, fanless box PCs	AMD G T40E	1GHz	2	2	GPU AMD Radeon HD 6250 (280MHz)	x86
PC compatible computer on modules, fanless box PCs	AMD G T56N	1.6GHz	2	2	GPU AMD Radeon HD 6310 (500 MHz)	x86
PC compatible computer on modules, fanless box PCs, HMI	Intel Atom Z510	1.1GHz	1	1	None	x86
PC compatible computer on modules, fanless box PCs, HMI	Intel Atom N455	1.66GHz	1	2	Integrated GPU (200MHz)	x86
PC compatible computer on modules, fanless box PCs	Intel Atom D525	1.8GHz	2	4	Integrated GPU	x86
PC compatible computer on modules, fanless box PCs, HMI	Intel Atom D2550	1.86GHz	2	4	Integrated GPU (640 MHz)	x86
PC compatible computer on modules, fanless box PCs, HMI	Intel Atom E680T	1.6GHz	1	2	Integrated GPU (400MHz)	x86
PC compatible computer on modules, fanless box PCs	Intel Core i7 3517UE	1.7GHz	2	4	Intel HD Graphics 4000 (350MHz)	x86
PC compatible computer on modules	AMD Geode LX800/ LX900	500/600 MHz	1	1	Geode LX Graphics/Video Processor, Geode Link Control processor	x86
Low-power com, HMI	Freescale iMx 6Solo	Upto 1GHz	1	1	NEON MPE coprocessor, GPU Vivante GC880(3D), GPU Vivante GC320(2D), IPU	ARM cortex A9
Low-power com, HMI	Freescale iMx 6Dual	Upto 1.2GHz	2	2	NEON MPE coprocessor, GPU Vivante GC2000 (3D), GPU Vivante GC320(2D), IPU	ARM cortex A9
Low-power com, HMI	Freescale iMx 6Quad	Upto 1.2GHz	4	4	NEON MPE coprocessor, GPU Vivante GC2000 (3D), GPU Vivante GC320(2D), IPU	ARM cortex A9
Low-power com, HMI	TI AM 3874	1GHz	1	1	NEON, SGX530 3D graphics engine	ARM cortex A8
Low-power com, HMI	TI AM 3517	600 MHz	1	1	NEON, POWERVR SGX Graphics Accelerator	ARM cortex A8
Fanless box PCs, HMI	Freescale iMx 537	800 MHz	1	1	NEON co-processor, VFP lite coprocessor, integrated GPU, IPU	ARM cortex A8
Fanless box PCs	Nano X2 E /L4350E	1.6GHz	2	2	none	x86
Fanless box PCs	Eden X2 /L4200E	1.0GHz	2	2	none	x86
Mobile devices, HMI, low-power com	Nvidia Tegra 3(T33)	1.6GHz	4	4	GPU ULP GeForce	ARM cortex A9
Mobile devices	Qualcomm Snapdragon 600	1.9GHz	4	4	GPU Adreno 320, DSP: Hexagon & QDSP6V4	ARM
Mobile devices	Intel Atom Z2760	1.8GHz	2	4	Integrated GPU (533MHz)	x86
Mobile devices	AMD A4 5000	1.5GHz	4	4	GPU AMD Radeon HD 8330 (500MHz)	Jaguar
Mobile devices	Mediatek MT8125	1.6GHz	4	4	GPU PowerVR SGX	ARM cortex A7
Gaming consoles(XBOX 360S)	XCGPU	3.2GHz	3	6	Xenos GPU	Power Architecture
Gaming consoles(PS3)	Cell Microprocessor	3.2GHz	1(PPE)	8(SPE)		Power Architecture

Human machine interface (HMI)

HMIs in industrial designs handle human-machine interactions. These can vary from simple switches, keypads to touchscreens and panel PCs. Most of the HMI solutions in the market are x86 based, using processors from Intel and other manufacturers that provide performance and flexibility. Industrial HMI requirements

are increasing the need for greater processing performance and more interface supports, and at the same time maintaining minimum heat generation and compact size, which is contributing towards use of ARM processors for HMI designs.

Future ahead

Every now and then, newer versions of chips are released, however when

the time comes for an upgrade, future expandability is a very important factor. Gordon Moore saw an end to processor speed growth when he said, "We can't exceed the speed of light," back in 1998. Rather than focussing on high speed, a smart engineer needs to focus on optimum speed, which can be sustained. ●

Pankaj V. is a technical journalist at EYF Gurgaon, while Dilin Arand is senior technical correspondent at EYF Bengaluru